

Question #1

Given training set is

Doc	words	Class
1	I like this movie	Positive
2	ordinary cast but great script	Positive
3	interesting plot average film	Negative
4	movie is interesting but long and slow paced	Negative
Test : 5	ordinary cast average movie	?

Required is $P(d_s)$. We are to use Laplace Smoothing. Here $|V|=18$, $\text{Count(Positive)}=9$
 $\text{Count(Negative)}=12$.

$$P(\text{Positive}) = \frac{N_p}{N} = \frac{9}{18} = 0.5$$

$$P(\text{Negative}) = \frac{N_n}{N} = \frac{9}{18} = 0.5$$

$$P(\text{ordinary} | \text{Positive}) = \frac{1+1}{9+18} = 0.074$$

$$P(\text{cast} | \text{Positive}) = \frac{1+1}{9+18} = 0.074$$

$$P(\text{average} | \text{Positive}) = \frac{0+1}{9+18} = 0.037$$

$$P(\text{movie} | \text{Positive}) = \frac{1+1}{9+18} = 0.074$$

$$P(\text{ordinary} | \text{Negative}) = \frac{0+1}{12+18} = 0.033$$

$$P(\text{cast} | \text{Negative}) = \frac{0+1}{12+18} = 0.033$$

$$P(\text{average} | \text{Negative}) = \frac{1+1}{12+18} = 0.067$$

$$P(\text{movie} | \text{Negative}) = \frac{1+1}{12+18} = 0.067$$

$$P(\text{Positive} | d_s) = 0.074 \times 0.074 \times 0.037 \times 0.074 \times 0.5$$

$$= \boxed{0.0000075}$$

$$P(\text{Negative} | d_s) = (0.033)^2 \times (0.067)^2 \times 0.5$$

$$= \boxed{0.0000024}$$

Hence, class prediction for d_s is Positive.

Question #2

(a)

Short documents (training sets) benefit more from the smoothing as they have limited (or less) number of words as compared to the long documents and hence the chances of getting unseen words in the test set increases thereby decreasing the efficiency of the language model. For large documents most words are already covered and smoothing does not improve the efficiency the way it does for small training data documents.

d)- Machine Translation

In natural languages, the meaning of the word is usually defined by its context. Translating the word itself doesn't always give the accurate answer. A word could have multiple translations based on the context. For example, avocado and advocate, both words are translated to "avocat" in French language. Both words have completely different meaning and can't be used interchangeably. Language model helps to determine which translation has higher probability / chance based on context.

Question # 3

(a)

Given three-document corpus after removing stopwords is

- d₁ information retrieval process index search retrieval
- d₂ retrieval used evaluation search results retrieval
- d₃ evaluation information evaluation process search

we know that

$$Tf = \begin{cases} 1 + \log(tf) & \text{if } tf > 0 \\ 0 & \text{otherwise} \end{cases}$$

$$IDF = \log\left(\frac{N}{df}\right)$$

For word information we get

$$tf = 0 \quad (\text{as this word does not exist in } d_2)$$

$$IDF = \log\left(\frac{3}{2}\right)$$

$$IDF = 0.176$$

Hence

$$\boxed{TF \cdot IDF(\text{information}) = 0 \times 0.176 = 0}$$

For word retrieval, we get

$$tf = 3, \text{ so}$$

$$TF = 1 + \log(3) = 1.477$$

$$IDF = \log\left(\frac{3}{2}\right) = 0.176$$

$$\boxed{TF \cdot IDF(\text{retrieval}) = 1.477 \times 0.176 = 0.26}$$

Q#3(d):

$$\text{cosine}(\text{fruit}, \text{Healthy}) = \frac{0 \times 3 + 0^{4 \times 0} + 6 \times 4 + 10 \times 3}{\sqrt{0^2 + 4^2 + 6^2 + 10^2} \sqrt{3^2 + 0^2 + 4^2 + 3^2}}$$

$$= \frac{0 + 0 + 24 + 30}{12.328 \times 5.83}$$

$$= 0.751$$

etc will occur in documents related to cricket

(C)

Given document is
he was driving the car on the road
and the car

The vector of such document would
be

< dimensions : weights >

< he : 1 >

< she : 0 >

< driving : 1 >

< the : 3 >

< car : 2 >

< was : 1 >

< road : 1 >

< on : 1 >

< and : 1 >

< fell : 0 >

< ground : 0 >

(6)

F-Score (using harmonic mean of P & R) gives a high value only if both P and R values are high. Harmonic mean has a property that it is inclined towards the small value and hence f-score value being low tells us that one of the values is low where arithmetic mean simply gives an average without giving any hint of whether the values are same or one of the value was small. Therefore, F-Score is used as we need both Precision and Recall to be good.

Question # 5

(a)

F-Score is a better measure as compared to accuracy as it tells about the precision and recall of the model. Some times (depending on the problem) we need good accuracy as well as good precision or good recall for our model. Accuracy alone is not enough to evaluate a model. For problems like cancer classification, we need high recall while we cannot compromise on precision too. Hence F-Score should be good.

Furthermore for class imbalanced data where one class dominates giving high accuracy we look at F-Score (Macro average) to understand how well has the model been trained.
